



***Sayed SAADE***

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SAHEL ALMA-Lebanon    Mobile: 70-190952

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*Client:*

*Dr. Ziad Saliba*

*Subject:*

*Specifications -1-*

*Project location:*

**FANAR 212**

## General Site protection Works

- Public safety: Protecting the site and maintaining safety by installing panels, partitions, and all necessary protective measures for internal and external work areas.
- Insurance for the project and workers throughout the work period, adhering to safety regulations.
- Insurance coverage for project and workers during the work period (according to emergency work insurance, Legislative Decree No. 136 issued in 1983).

## Concrete Works

- Concrete strength: 25–30 MPa according to ACI (American Concrete Institute) International Code.
- Main Steel reinforcement: Yield strength  $F_y = 4200 \text{ kg/cm}^2$
- Designed to meet international seismic requirements
- Foundations, tie beams, and columns: Concrete strength = 30 MPa.
- Slabs and beams: Concrete strength = 25 MPa.
- Execution of slabs, columns, tie beams, and blinding layers with double nylon sheets between blinding and natural soil, according to structural drawings. Includes vibration, tying steel, raising it off wood, watering slabs before and after pouring twice daily for 7 days.
- Final slab includes rainwater drainage toward outlets.
- Lightweight reinforced concrete: Minimum compressive strength 25–30 MPa after 28 days.
- Plain concrete (blinding layers): Minimum compressive strength 15 MPa after 28 days.
- All concrete must remain moist for 28 days.

## Masonry Works

- Internal walls: 10 cm concrete masonry unit (CMU) walls
- External walls: Double 10 cm CMU walls with a 5 cm cavity for thermal and acoustic insulation
- Internal walls in bathrooms: 20 cm block to align with columns.
- Use GL steel every 4 layers; GL angle 4 mm where needed.
- Reinforced lintels above openings with 30 cm overlap.
- Cast reinforced concrete lintel (10 cm height  $\times$  25 cm width) at stairwell walls to prevent water leakage.
- For walls over 350 cm high, add an intermediate reinforced tie beam.

## Plastering Works

- Internal plastering: 3 m bar height, for walls and ceilings.
- External plastering: On walls, all layers mix to include bonding agent. (Poly Bond)

Process:

1. First coat: rough spray, watered for 3 days.
2. Second coat: rough plaster, watered for 3 days. (For cladding facades)
3. Final coat: smooth plaster, watered for 3 days. (For painted facades)
4. Mesh installed at joints between walls and columns.
5. All bathrooms wall.

## Waterproofing

- Adequate waterproofing for all concrete in touch with the underground soil protecting the building foundation from erosion and steel reinforcement from rust using a single layer of hot applied SBS elastomeric membrane.
- For basement walls a double layer of hot applied SBS elastomeric membrane 4mm thick is installed, protected by a 10 cm concrete masonry unit to prevent future damages.
- Follow this sequential approach to ensure the integrity of each layer:
  - Substrate Preparation
    - Smooth and clean the slab to remove all debris, dust, and loose particles. Use a concrete polisher or grinder to achieve an even surface.
  - Waterproofing Application
    - Primer: Apply a coat of bituminous primer to seal the pores and allow for maximum adhesion.
    - Membrane (2 coats): Install two full coats of 4mm thick SBS-modified bituminous membrane, reinforced with (180 gr/m<sup>2</sup>) polyester. The membrane should be hot-applied (torch-on method) with a propane torch, ensuring a minimum of 10cm side laps and 15cm end laps.
  - Thermal Insulation
    - Lay down 3/5 cm thick extruded rigid cellular polystyrene (XPS) boards. Stagger the joints to prevent thermal bridging and ensure a continuous vapor and thermal barrier.
  - Screeding
    - Light Reinforced Screed: Pour a 6 cm light reinforced screed over the XPS boards.
    - Preparation: Ensure the screed is perfectly leveled, smoothed, cleaned, and given enough time to cure, leaving the surface ready for final tile installation.
- For water tanks and pools: Cementitious based elastic waterproofing with fiber glass reinforced mesh, alkali resistant is applied.
- For flower beds and wet areas: 2 coats of PU (polyurethane) granting zero leak to beneath floor, and using a light reinforced screed for bathrooms floor.
- Waterproofing around all drains.

- Apply two coats of bituminous waterproofing coating to exterior walls behind wood cladding.
- Apply 2 layers of cementitious waterproofing coating with reinforced mesh around windows and under copings.

## Mechanical Works

- European polypropylene for embedded pipes inside the units.
- European galvanized steel pipes for circulation and common areas.
- UPVC pipes for drainage and sewage.
- 1 Hot water tank for each unit of 200L capacity.
- For each unit, 1 water tank of 2500L in basement level with European pump leading to private tank of 1000 L on the roof top level. On Attic level, booster pump included only for the last 2 floors.
- All WC are wall hung toilet.
- Gully traps installed in any drainage line start to ensure smooth smell and ventilation system including the vent pipes and P-trap, floor traps and fittings s-traps to ensure complete smell lightness.
- For Kitchens, provide only an exhaust with 6" flexible and a20x20 aluminum grill. No dedicated fan is provided. Customers will connect their hood to the 6" flexible.
- Fixtures: High quality brand glossy white. *Ideal Standard* or equivalent.
- Faucets: *Grohe*.
- White PVC, slim shower trays of 80x80 or 120x80 with standard light sliding closure.

## Heating and Cooling Works

- AC wall type inverter system MDV white or equivalent  
Bedrooms 9000 BTU, Living rooms 18000 BTU.
- Heating system using Gas boiler to be installed on kitchen's balcony. (Optional)

## Electrical Works

- Earthing protection system for all equipment and sockets outlets.
- Wiring devices: 4 mm (Liban cables)
- Lighting circuits: 2 mm (Liban cables)
- Telephone wiring installed (1 outlet for each unit)
- TV wiring installed (4 outlet for each unit)
- Data CAT 6 wiring installed (1 outlet for each unit)
- Main source: 10 mm & above as required by design (Liban cables)
- All wires to be encased in a smooth flex tubing.

- Wiring accessories: Le Grand, Arteor, White or equivalent.
- Electric Panel Boards and normal breakers: Schneider or equivalent.
- Video phone system to be installed at the main gate, Urmet or equivalent.
- Reservation for home Automation (Optional)
- Solar panel provision for common area (Only panels and wiring to be installed)

## Façade Finishing

- Cementitious based paint for exterior façade, providing best solutions for water leakage with layers of high-quality exterior paint to give ultimate modern effect and beauty as per the displayed perspectives.
- External cladding: Compact laminate wooden effect cladding mechanically fixed with distinguished durability and weather resistance (UV resistance – chemical resistance and warrant for any change in color).
- Marble or natural stone cladding to be installed if required legally. (To discuss fixation and insulation)

## Wood Works

- Entrance door: Solid wood door of 4.5 cm thick, solid frame of oak wood or equivalent, with European hardware & fittings Yale or equivalent.
- Interior door: White painted green MDF pressed door with Chinese or Turkish black hardware and fittings.

## Tiling Finishing

- Living room, entrance and balcony: Premium Full body ceramic tile not less than 60 x 120.  
Marble texture finishing, European source, Polished or matte (As per design selection)
- Bedrooms and corridors: Premium Full body ceramic tile not less than 20 x 60.  
Wooden texture finishing, European source, Polished or matte (As per design selection)
- Kitchens and bathrooms: Premium Full body ceramic tile not less than 30 x 90.  
Marble texture finishing, European source, Polished or matte (As per design selection)
- Bathroom floors: Waterproofed with PU, nylon sheets, 20 cm wall height, all shower area walls, floor screeding.
- Bathroom walls: Premium Full body ceramic tile not less than 20 x 60.  
Marble texture finishing, European source, Polished or matte (As per design selection)  
Tiled using European glue.

- Internal stairs: Granite or marble.
- Main entrance: Granite or marble.
- Outdoor terraces: European 60×60 tiles, moisture and slip resistant.
- Parking floor, to be stamped concrete finishing.
- Storage area and basement technical rooms: Full body ceramic tile not less than 60 x 60.

## Kitchen

- Electrical and Sanitary connections for all appliances provided.
- Kitchen walls white painted.
- Earthing and leakage protection for main kitchen outlets, ensuring safety when using appliances such as the refrigerator, washing machine and dishwasher.
- Kitchen cabinets: Egger Laminated wood, Blum accessories and drawers.  
Countertops with backsplash: Granite Galaxy.

## Painting Works

- Interior Painting: Prepare and apply one coat of primer sealer, undercoat, 2 coats of putty and 3 coats of paint for walls and ceilings, Noula or equivalent, are applied to ensure smooth high-end finishing.
- Exterior painting: Cementitious based paint, providing best solutions for water leakage with layers of high-quality exterior waterproofed paint, colors as per design. (Monocouche/ Grattin)
- Main entrance wood doors: Sanding, putty, furniture spray paint.
- Iron main gate: Sanding, putty, primer, spray paint.

## Aluminum & Glazing

- Doors and windows Aluminum: Local brand Reina or equivalent, color as per design.
- Doors and windows Glass: double glazing in 6/9/6 mm, clear color.
- Mosquito net (Optional)
- Balconies balustrade to be tempered glazing with aluminum frame.
- Balconies closure to be **Included or not? Without internal aluminum doors?**
- Shower tray tempered glazing doors (Optional)

## False ceiling

- Bathrooms ceilings: PVC false ceiling panels.
- Balconies ceilings: PVC false ceiling panels.
- Gypsum cornices in main entrance and living room, with hidden lighting or embedded magnetic track lighting (Optional).

## Lift

Product Specifications: Made in Italy, Room less.

- Motor Power: 6 HP, dual-speed electronic system
- Accessories: All included, made in Italy
- Warranty & Maintenance:
  - 1-year full warranty and maintenance by the company from delivery date
  - 5 years warranty on the motor
  - 2 years warranty on the electronic control panel
- Additional Feature: Equipped with UPS for power backup in case of electricity outage.

## Fiber Optic

- Fiber cables will connect each unit to the main electrical room.
- Following the US & the EU standards to guarantee a high speed and quality experience.
- Benefits of high-speed internet, High quality HD cables TV, High quality phone lines

## Roof Tiles

- Concrete roof solid slab, well waterproofed isolated.
- Galvanized steel structure attic above, add a thermal insulation layer of 5 cm extruded panels.
- Red European roof tiles, per architectural design.
- Solar panels installed between brick for common area.

## Iron Works

- Main hall gate: Decorative iron + glass, European locks (Yale), intercom.
- External fences: Iron. (Where needed) with anti-rust epoxy primer + paint.

## External Works

- Landscaping: Minimum 10 trees per garden,  $\geq 2$  m height.
- Road asphalt, parking areas stamped concrete.
- Rainwater drainage, sewage system.
- External and covered parking lighting.
- Main building entrance false ceiling design and decorative lighting.